



How Does the Downstream Accuracy of Barlow Twins Scale with Pre-training Set Size?

A small-compute characterization with a ViT-Tiny on Tiny-ImageNet subsets

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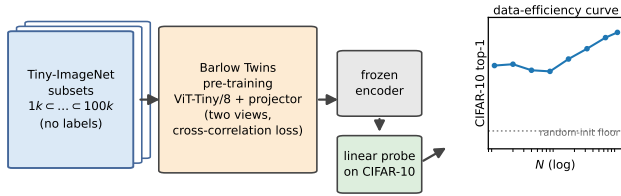


Figure 1: The data-efficiency pipeline. Barlow Twins pre-trains a ViT-Tiny encoder on an unlabeled Tiny-ImageNet subset of size N ; a linear classifier trained on CIFAR-10 scores the frozen encoder. Repeating this across N yields the curve we study (right panel: the measured seed-42 curve of Section 4.2): how much downstream accuracy each amount of unlabeled data buys.

Abstract

Self-supervised pre-training works best with millions of images and the compute to match, which puts visual foundation models out of reach for most groups. We characterize how one widely used self-supervised method, Barlow Twins, behaves when both data and compute are small. A 5.4M-parameter vision transformer is pre-trained on subsets of Tiny-ImageNet (1k to 100k images); each resulting network, its weights then frozen, is scored by training a linear classifier on CIFAR-10 — a downstream task the network never saw. Checkpoints are selected with a validation score computed only on the pre-training dataset, never on the downstream task. In the completed single-seed campaign, pre-training pays at every set size: 61.4% downstream accuracy from only 1k unlabeled images and 71.5% from 100k, against a 41.5% floor for an untrained network. The central finding is that the pre-training validation score and downstream accuracy disagree about *which checkpoint is best*: downstream-blind selection picks checkpoints that score up to 4.7 points worse on a CIFAR-10 monitoring metric than the best checkpoint in hindsight. The disagreement peaks at the mid-range set sizes, where the measured curve even dips below its 1k value — the dip itself lies within the seed variance of an earlier campaign and awaits confirmation; the divergence does not. On the smallest set, validation accuracy decays 18% from its peak with no sign of representational collapse. At small scale, the rule for when to stop and which checkpoint to keep shapes a data-efficiency curve alongside the data itself.

1 Introduction

Pre-training a visual foundation model [1] is expensive twice over: it takes millions of images and the compute to match [2, 12]. Self-supervised learning (SSL) removes the annotation cost by training image representation models on unlabeled images [4, 28], but the appetite for data and compute remains, so groups without either — university labs, small companies

— can rarely train or even study these models. Making SSL useful in that setting requires understanding how SSL methods behave when both the pre-training dataset and the compute budget are small. This work characterizes one widely used SSL method, Barlow Twins [28], in exactly that regime.

We study the question along the data axis. The *data efficiency* of a pre-training method is how its downstream accuracy grows with the number of unique pre-training images N , and N is the only variable we manipulate. Training length is not held equal across set sizes. Instead, every set size trains under a generous fixed epoch budget, and we evaluate the checkpoint selected by an in-domain validation signal — one computed on the pre-training dataset — that never sees the downstream task. Each curve point is therefore read at its best validation point within its budget, which is what makes points at different N comparable (Section 3). These are best-within-budget readings, not converged ones.

Figure 1 shows the experimental pipeline. A small vision transformer (ViT-Tiny with 8×8 patches, 5.4M parameters) [7, 26] is pre-trained with Barlow Twins on subsets of Tiny-ImageNet [17] ranging from 1000 to 100 000 images. The encoder is then frozen — its weights no longer update — and a linear classifier is trained on CIFAR-10 [16]; its test accuracy — the *linear-probe accuracy* — measures the quality of the learned representation on a downstream task the encoder never saw. This pipeline is a deliberately scaled-down version of the foundation-model recipe: pre-train once on broad data, adapt cheaply to a downstream task. The study is part of a five-student project in which each student applies a different SSL method to this shared pipeline (same backbone, same data subsets, same probe); we chose Barlow Twins, and this paper characterizes its behavior.

Barlow Twins is an interesting subject for a small-data study. It learns by pushing the cross-correlation matrix between the embeddings of two augmented views of the same images toward the identity matrix, which makes representational collapse unattractive by construction, without the negative pairs of contrastive methods [4] or the asymmetric momentum/predictor networks of self-distillation methods [2, 10]. At the same time, the joint-embedding family it belongs to has been reported to depend more strongly on large pre-training sets than masked-prediction alternatives [8]. Small-data SSL in general is well studied (Section 2); Barlow Twins specifically, at this small a scale and on a small ViT, is not — its small-data behavior is both scarcely described and, given its family’s reported data appetite, not obvious in advance.

We therefore ask: *how does the downstream accuracy of Barlow Twins scale with pre-training set size for a small vision transformer in a small-compute regime, and how does it behave at the smallest scales?*

Our contributions are:

- The Barlow Twins data-efficiency curve (Tiny-ImageNet $\rightarrow$ CIFAR-10 linear probe) for a ViT-Tiny across $N \in \{1k, \dots, 100k\}$, reported against a random-initialization baseline — including the observation that, under our fixed generous budgets, the curve is *not* monotone in N .

- An examination of the smallest-data regime: Barlow Twins trains past its validation peak and declines, while the effective rank [15,21] of the representation — a measure of how many embedding dimensions carry variance — stays high: the decline is not dimensional collapse.
- A quantified divergence between the in-domain selection signal and downstream accuracy: selecting checkpoints downstream-blind costs up to 4.7 diagnostic points against a downstream oracle, and the cost peaks at exactly the set sizes where the curve dips.

All findings are observational and scoped to this setup: one backbone, one pre-training dataset, one downstream task, and — in this draft — one completed pre-training seed (Section 6).

2 Related Work

Most self-supervised methods for images learn representations by making the embeddings of two augmented views of the same image agree, while avoiding the trivial solution in which every image maps to the same embedding (complete collapse). Contrastive methods such as SimCLR [4] and MoCo-v3 [5] prevent collapse by also pushing apart embeddings of different images. Self-distillation methods such as BYOL [10] and DINO [2] prevent it with asymmetry between two networks, and masked autoencoders [12] avoid view agreement altogether by reconstructing masked input patches. Barlow Twins [28] instead pushes the cross-correlation matrix between the two views’ embeddings toward the identity matrix: the diagonal enforces view invariance, the off-diagonal decorrelates embedding dimensions, and the objective itself penalizes collapse with neither negative pairs nor network asymmetry. We adopt the Barlow Twins objective and its reference loss implementation unchanged, adapt the recipe to 64×64 inputs and small-batch ViT training (Section 3), and characterize how the method behaves as pre-training data shrinks.

How much pre-training data self-supervised learning needs is a well-studied question. Contrastive learning has been measured across data quantity, domain, and quality, with modest benefit from pre-training data beyond 500k images [6]. Closer to our scale, denoising autoencoders (BEiT-style masked prediction) have been reported more robust to the size and type of the pre-training set than joint-embedding methods, represented there by DINO and evaluated by fine-tuning; under linear probing, the same study reports denoising autoencoders falling behind [8]. At the opposite end of the scale, joint-embedding training on a large enough dataset needs almost no augmentations beyond cropping: hand-crafted augmentations act chiefly by artificially enlarging the dataset, so their benefit concentrates where data is limited [20] — though the smallest scale tested there is 1.3M images, far above ours. These studies chart the small-data regime across method families, but none of them measures Barlow Twins. So we characterize the data efficiency of Barlow Twins directly, on a small ViT, from 1k to 100k pre-training images, under a linear-probe evaluation.

Comparing models pre-trained on different set sizes requires a schedule that does not tie model quality to a pre-

committed stopping point, and cosine learning-rate schedules [18] do exactly that: a checkpoint read out well before the cosine horizon attains higher loss than a model whose schedule was matched to that duration [13]. Warmup-stable-decay (WSD) schedules avoid the committed horizon and match the final loss of a well-tuned cosine [11, 14]; horizon-free schedules of this family have precedent in vision-transformer scaling [29] and continual pre-training [22]. So we train every set size with a WSD-style schedule, assembled from these published components (Section 3); the schedule is not a contribution of this work.

Self-supervised training also needs a validation signal for checkpoint selection. The standard monitor is k -nearest-neighbor classification on frozen features over held-out in-domain data, as in DINO with $k=20$ [2]; fully label-free alternatives based on embedding rank exist [9], which we log but do not select with. So we follow the k NN convention, smoothed against selection bias [3] as Section 3 details.

A representation can also degrade without collapsing completely: in dimensional collapse, the embedding variance concentrates in a low-dimensional subspace [15], which the effective rank of the feature covariance spectrum quantifies [21]. Augmentations are not a free remedy in this regime. The invariances a joint-embedding method learns are the invariances its augmentations define [27], and views that share too little information destroy task-relevant signal [25]; for supervised ViTs, augmentation and regularization can substitute for an order of magnitude of training data [23], underlining how strongly augmentation choices act as implicit data. So in the small-data regime we keep the augmentation recipe fixed — the published one, adapted once to 64×64 inputs (Section 3) — and track effective rank as a logged collapse diagnostic.

3 Method

The method has four parts: a small vision-transformer encoder (Section 3.1), the Barlow Twins objective on two augmented views (Sections 3.2 and 3.3), and a training schedule with downstream-blind checkpoint selection (Section 3.4, Figure 2). The encoder and objective follow their published versions, and the schedule is assembled from standard, cited components. The recipe deviates from published defaults in a handful of places — λ , projector width, two augmentation bounds, the optimizer — each marked where it occurs; the untested constants among them are collected in Section 6.

3.1 Encoder and projector

The encoder is a ViT-Tiny [7,26] with 8×8 patches operating on 64×64 images: 12 transformer layers, 192-dimensional embeddings, 3 attention heads, 5.4M parameters. The patch size matches the native resolution of the pre-training images, giving $8 \times 8 = 64$ patch tokens plus a classification (CLS) token. The representation evaluated downstream is the CLS token after the final layer normalization. During pre-training only, a projector maps this 192-dimensional embedding to the space where the loss is computed: three linear layers ($192 \rightarrow 1024 \rightarrow 1024 \rightarrow 1024$), the first two followed by batch normalization and ReLU. The projector follows the

Barlow Twins design [28] scaled down from its original 8192 dimensions, at which the projector alone would exceed the encoder in parameter count. It is discarded after pre-training.

3.2 Barlow Twins objective

Barlow Twins makes the projector outputs of two augmented views of the same images agree, while making the individual output dimensions carry **non-redundant** information. Both **pressures** act on one matrix: the cross-correlation between the two views’ outputs, computed across the batch. With $z^A, z^B \in \mathbb{R}^{B \times D}$ the batch-normalized projector outputs of the two views (B the batch size, $D = 1024$ the projector width), the cross-correlation matrix $C \in \mathbb{R}^{D \times D}$ has entries

$$C_{ij} = \frac{\sum_b z_{b,i}^A z_{b,j}^B}{\sqrt{\sum_b (z_{b,i}^A)^2} \sqrt{\sum_b (z_{b,j}^B)^2}}, \quad (1)$$

where b indexes the batch and i, j index output dimensions. The loss pushes C toward the identity matrix,

$$\mathcal{L}_{\text{BT}} = \sum_i (1 - C_{ii})^2 + \lambda \sum_i \sum_{j \neq i} C_{ij}^2, \quad (2)$$

where the first term enforces invariance across views, the second decorrelates output dimensions, and λ weights the two. A constant representation cannot satisfy Equation 2: the diagonal of C requires unit per-dimension correlation across views, so the objective itself penalizes complete collapse, without negative pairs or asymmetric networks [28]. We use the reference implementation of the loss from the `lightly` library [24] rather than reimplementing it, because the loss has known silent failure modes — normalization along the wrong axis, incorrect off-diagonal scaling — that a reference implementation rules out. We set $\lambda = 10^{-2}$; the reference default $5 \cdot 10^{-3}$ was tuned at projector width 8192 [28].

3.3 View generation

The two views are produced by the published two-view BYOL augmentation pipeline [10], as implemented in `lightly` [24]: random resized crop, horizontal flip, color jitter (strength 1.0), and random grayscale, with the second view additionally solarized with probability 0.2. Two parts of the recipe are adapted to 64×64 inputs. Gaussian blur is disabled on both views, since its kernel is sized for 224×224 images. The random crop retains at least 25% of the image area rather than the published 8%. The crop scale is an area fraction, so at 64×64 an 8%-area crop keeps roughly 18×18 pixels — about 2×2 of our 8-pixel patches — and two such views may share almost no content; views that share too little information lose task-relevant signal [25]. At 224×224 , where the recipe was tuned, the same fraction still keeps roughly 63×63 pixels.

3.4 Training schedule and checkpoint selection

Training follows a warmup-stable-decay (WSD) schedule [11, 14, 29] (Figure 2). The learning rate warms up linearly over the first 500 optimizer steps, then stays constant for the remainder of a fixed epoch budget. We warm up in optimizer steps rather than the conventional “10 epochs” [2, 28]

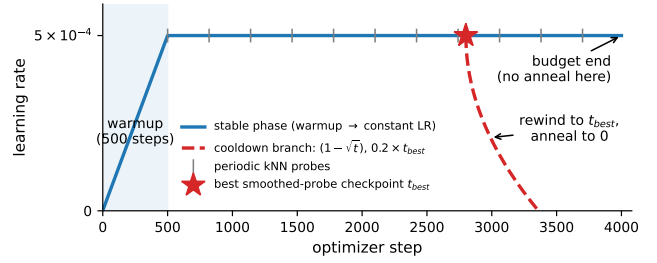


Figure 2: The training schedule, drawn with the 1k-split geometry; the learning-rate curves are the exact functions used in training. The rate warms up for 500 steps and stays constant while a kNN probe periodically scores the encoder; after the budget ends, training rewinds to the best-probe checkpoint for a short $(1 - \sqrt{t})$ cooldown. No training horizon is committed in advance.

because warmup stabilizes optimizer updates, and an epoch-based rule degenerates across our set sizes: at batch 250 it would give 40 warmup steps on 1k images and 4 000 on 100k.

A constant rate, rather than the cosine decay common in self-supervised pre-training, serves the selection mechanism. Under a cosine schedule, a checkpoint read out well before the horizon attains higher loss than a model whose schedule was matched to that duration [13, 18], so checkpoints from different training stages are not comparable. Under a constant rate they are, and runs stay extendable because no horizon is baked into the schedule. The convenience costs nothing in final loss at language-model scales [11]; whether that finding transfers to our setting is an assumption (Section 6).

During the stable phase, a k -nearest-neighbor probe on held-out in-domain data scores the encoder at a fixed epoch interval (Section 4.1). The checkpoint with the best smoothed probe accuracy, the mean of the last three probes, is retained. Smoothing matters because the maximum of a noisy, repeatedly evaluated signal is biased toward flukes [3]. The probe never sees the downstream dataset; selection is in-domain, following standard practice [2].

After the budget completes, a cooldown branch restarts from the selected checkpoint and anneals the learning rate to zero over $0.2 \times t_{\text{best}}$ steps with a floor of 500, where t_{best} is the selected checkpoint’s step. The $(1 - \sqrt{t})$ decay shape attained the lowest final loss in the cooldown-shape ablation of Hägele et al. [11], and decays of roughly 10% of training suffice where 2.5% falls short [14]. Branching the cooldown from a checkpoint follows MiniCPM [14]. Rewinding to the *best-validation* checkpoint rather than the last one is our own adaptation: in preliminary runs on the smallest split, probe accuracy peaked and then declined within the budget, and annealing the last checkpoint would anneal a degraded model. The annealed model is the one evaluated downstream.

Throughout, we train with AdamW [19], weight decay 10^{-4} and gradient-norm clipping 1.0, at batch size 250. AdamW replaces the original recipe’s LARS optimizer, which targets large-batch convolutional training; at our batch size, ViT self-supervised methods use AdamW throughout [2, 5, 12]. The batch size is 250 rather than the conventional 256 because 250 divides every pre-training set

size, so with a drop-last loader one epoch is exactly one exposure of every image at every N — with batch 256, an epoch on the 1k split would silently skip almost a quarter of it. The fixed epoch budget is generous rather than calibrated: we do not claim the selected checkpoints are trained to convergence, and Section 4.2 reports per set size whether the validation signal plateaued within its budget.

4 Experimental Setup and Results

4.1 Experimental setup

Pre-training data and splits. Pre-training uses Tiny-ImageNet [17]: 100 000 training images, 200 classes, native 64×64 resolution (Figure 3); labels are never used during pre-training. The pre-training set sizes are $N \in \{1k, 2k, 4k, 8k, 16k, 32k, 64k, 100k\}$, drawn as prefixes of a single shuffled master list (seed 42) that interleaves one image per class, so every split is exactly class-balanced and smaller splits are subsets of larger ones. The nesting means a larger split differs from a smaller one only by the images added, so curve differences are attributable to N alone.

Budgets and validation probe. Each split trains a fixed epoch budget of $\{1k\text{--}8k: 1000, 16k: 600, 32k: 400, 64k: 300, 100k: 250\}$ epochs, sized from pilot runs to lie generously past the validation peaks observed there; budgets shrink with N because small splits peak only after hundreds of epochs, while large splits peak within tens to hundreds. The budget is therefore a second protocol variable that varies with N — total optimizer steps span a $25\times$ range across the curve — which the best-within-budget reading is designed to absorb, and which Sections 4.4 and 6 revisit. Table 2 reports where each split actually peaked. The validation probe runs every four epochs on the three smallest splits, every two epochs on 8k and 16k, and every epoch from 32k upward, so that larger splits — which run fewer epochs — still receive several hundred probes. The Tiny-ImageNet valid split — 10 000 labeled images held out from all pre-training splits — is partitioned class-stratified into a 5 000-image gallery and a 5 000-image query set, 25 per class each. Query images are classified by their $k=20$ nearest gallery neighbors in CLS-embedding space, with Euclidean distance and no weighting. The same probe records the effective rank of the query embeddings (the dimensional-collapse diagnostic of Section 4.3) and, as a logged-only diagnostic that never influences selection, a k NN accuracy on CIFAR-10 (Section 4.4).

Downstream evaluation. The annealed encoder is frozen, and a single linear layer ($192 \rightarrow 10$) is trained on CIFAR-10 [16], resized $32 \rightarrow 64$ (bilinear) and ImageNet-normalized to match pre-training statistics. The probe protocol is shared across the five-student project: SGD with momentum 0.9, no weight decay, batch 512, learning rate 0.2 ($0.1 \times \text{batch}/256$, the linear scaling rule), cosine annealed to zero over 100 epochs, fixed evaluation seed independent of the pre-training seed. The training transform includes a random horizontal flip. The reported number is the best test top-1 across the 100 probe epochs; the shared evaluation script does not yet persist the final-epoch accuracy, so the size of this inflation is not reported in this draft (Section 6).

Table 1: Learning-rate ablation on the 8k split (single seed, 500 epochs). Accuracy is the in-domain kNN top-1 at the selected checkpoint; effective rank (of 192) is measured at the budget end. 5×10^{-4} sits mid-band with near-peak effective rank, and 1.5×10^{-3} fails outright.

Learning rate	kNN top-1 (%)	Effective rank
1×10^{-4}	13.9	55
3×10^{-4}	14.7	100
5×10^{-4}	15.3	94
7×10^{-4}	15.8	81
1.5×10^{-3}	3.9	8

Tiny-ImageNet (pre-training, 64×64 , no labels used)



CIFAR-10 (downstream probe, 32×32 shown resized to 64×64)

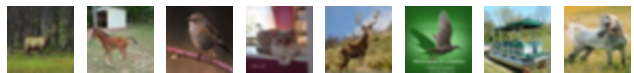


Figure 3: Pre-training and downstream data: Tiny-ImageNet (top, 64×64) and CIFAR-10 (bottom, resized as in the probe pipeline). The datasets come from disjoint source collections, so the probe measures transfer of the representation, not recall of the pre-training set.

Baselines. A random-initialization floor applies the identical probe protocol to an untrained encoder; it isolates how much of the linear-probe accuracy is bought by pre-training rather than by the architecture and the probe itself.

Seeds, tuning, and provenance. Every pre-training run is indexed by code commit, split size, and seed. The protocol fixes three pre-training seeds (42, 43, 44); this draft reports the completed seed-42 campaign, and the remaining seeds are in progress (Section 6). One hyperparameter was tuned: the learning rate, by a one-off single-seed ablation on the 8k split — chosen as a mid-range size — at an earlier revision of the training code (Table 1). Accuracy rises from 10^{-4} , which under-trains, to a usable band at $3\text{--}7 \times 10^{-4}$, then fails outright at 1.5×10^{-3} : that run never improves past its warmup phase and ends with an effective rank of 8 out of 192. Within the band the accuracy differences are about one point, while effective rank already declines beyond 3×10^{-4} . We picked 5×10^{-4} — mid-band, near-peak effective rank, and the reference value of DINO at batch 256 [2] — and use this single rate for all splits. All remaining hyperparameters are fixed by the shared protocol or by the published recipes of Section 3. All runs execute on one consumer GPU; Section 5 reports wall-clock costs.

4.2 How does downstream accuracy scale with pre-training set size?

Figure 4 shows the data-efficiency curve for the completed seed-42 campaign; Table 2 gives the per-split readings. Pre-training helps at every N : the smallest split already reaches 61.4%, 19.9 points above the random-initialization floor of 41.5%, and the largest reaches 71.5%, 30.0 points above it.

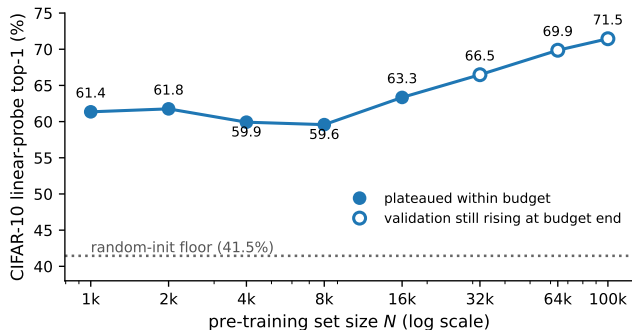


Figure 4: Data efficiency of Barlow Twins (seed 42, fixed budgets): CIFAR-10 linear-probe top-1 against pre-training set size N , with the random-initialization floor dotted; open markers are splits whose validation was still rising at budget end. Pre-training helps at every N , but the curve dips at 4k–8k: under this protocol, more unlabeled data does not monotonically improve transfer.

Table 2: Per-split protocol readings (seed 42). “Best” is the epoch of the selected (best smoothed in-domain validation) checkpoint; splits whose best epoch falls within the final probes of the budget had not plateaued. The selection signal (TI kNN) rises monotonically with N while the linear-probe accuracy does not.

N	Budget	Best	TI kNN (%)	Probe (%)
1k	1000	472	10.0	61.4
2k	1000	408	11.2	61.8
4k	1000	472	12.6	59.9
8k	1000	880	15.5	59.6
16k	600	522	17.5	63.3
32k	400	394	19.7	66.5
64k	300	288	22.8	69.9
100k	250	248	25.2	71.5

The curve is, however, *not* monotone in N under our protocol: the 4k and 8k splits score 59.9% and 59.6%, below the 1k split trained on a quarter of their data. From 16k onward accuracy rises steadily. The three largest splits selected their checkpoint within the final probes of their budget (best epoch 394 of 400, 288 of 300, and 248 of 250) — their validation signal had not plateaued, so their curve points are lower bounds under this budget. The dip at 4k–8k coincides with a measurable divergence between the selection signal and downstream accuracy, which Section 4.4 analyzes.

4.3 At the smallest scale, does Barlow Twins over-train — and is it collapse?

Yes, it over-trains; no, the decline is not dimensional collapse (Figure 5). On the 1k split, the smoothed validation accuracy peaks at 10.0% at step 1888 (epoch 472 of the 1000-epoch budget) and falls to 8.2% by the budget end — a relative drop of 18%. The effective rank of the embeddings tells the opposite story: it rises throughout training, from single digits during warmup to 103 of 192 at the budget end, and never falls (192 would mean every embedding dimension carries equal variance; a value near 1, a near-constant output). What-

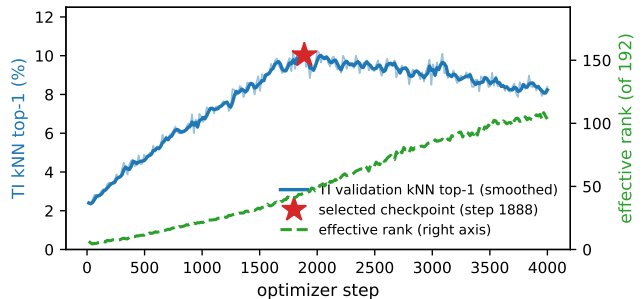


Figure 5: Over-training on the 1k split (seed 42): in-domain validation accuracy (left axis, raw and smoothed) and effective rank (right axis) against training step. Validation peaks at step 1888 and declines by 18% relative, while effective rank rises monotonically to 103 of 192: the late-training decline is not dimensional collapse.

ever degrades the representation late in training, it is not the variance collapsing into few dimensions. We read the decline as overfitting to the 1 000 pre-training images — an interpretation; we did not isolate the mechanism, and the constant learning rate is a candidate alternative (Table 1 shows accuracy is sensitive to it, and it was tuned on the 8k split, not here). The best-validation rewind exists for exactly this case: the selected checkpoint predates the decline by design.

4.4 Does the in-domain selection signal track downstream accuracy?

Across set sizes, yes; within a training run, increasingly no — and the mismatch peaks exactly where the curve dips. The selection signal itself rises monotonically with N (10.0% to 25.2% TI kNN, Table 2), so N ranks identically under either signal at the extremes; the disagreement concentrates at 4k–16k.

Figure 6 quantifies the within-run mismatch using the logged CIFAR-10 kNN diagnostic, which scores every probed checkpoint of every run but never influences selection. For each split we compare the diagnostic at the checkpoint our protocol selects against the diagnostic at its own (smoothed) peak — a downstream oracle unavailable to a downstream-blind protocol. The gap forms an inverted U over N : below two points at 1k and 2k, peaking at 4.7 points at 8k, and vanishing at the largest splits. Under the oracle the diagnostic improves monotonically with N ; under in-domain selection it dips at 4k–8k, mirroring the linear-probe curve. The 8k split is the extreme case: its in-domain signal kept improving until epoch 880 (step 28 160), while its CIFAR diagnostic had peaked at step 4 416 — the selected checkpoint lies 24 000 steps past the transfer optimum. These are diagnostic-level numbers; the linear-probe-level oracle comparison requires probing the diagnostic-peak checkpoints, which the pipeline now saves (Section 8).

Two further observations support reading the dip as a selection-and-budget effect rather than a property of the data quantity. First, at a matched step count (the probe nearest step 4 000, before any divergence), the diagnostic orders the mid splits normally: 44.9% (1k), 47.6% (2k), 52.2% (4k), 54.2% (8k). Second, an earlier three-seed campaign, run with

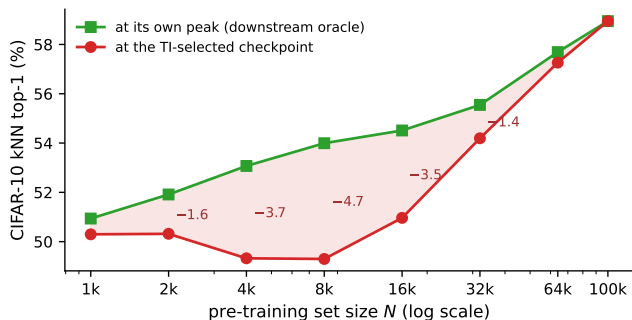


Figure 6: Cost of downstream-blind selection (seed 42; note the metric is the CIFAR-10 kNN *diagnostic*, not the linear probe of Figure 4): the diagnostic at the checkpoint selected by the in-domain signal versus at its own peak (downstream oracle). Under the oracle, more data monotonically helps; under in-domain selection a gap of up to 4.7 points opens at the mid splits — at the diagnostic level, the dip of Figure 4 coincides with selection-and-budget effects, not with the data quantity itself.

different budgets and with checkpoints selected by the CIFAR diagnostic itself, reached 64.4% and 65.7% mean linear-probe accuracy at 4k and 8k — 4.5 and 6.1 points above the downstream-blind protocol at the same splits. That comparison changes two variables at once (selection signal and budget), so it corroborates rather than isolates the effect. Our interpretation, labeled as such: with a generous budget, continued training keeps improving in-domain structure well past the point where the improvement transfers; at mid N the in-domain signal rises long enough for the transfer optimum to be left far behind, while at small N the signal itself declines early (rewinding selection toward the optimum) and at large N the budget ends first.

5 Discussion

This work trained Barlow Twins on nested Tiny-ImageNet subsets under fixed generous budgets with downstream-blind checkpoint selection, and measured each selected encoder by a CIFAR-10 linear probe against a random-initialization floor. Three observations stand out: pre-training pays at every set size (19.9 to 30.0 points over the floor), the curve dips at 4k–8k, and the dip coincides with the largest measured divergence between the selection signal and downstream accuracy.

What the dip means. The naive reading of a data-efficiency curve is that each point reflects the value of that much data. Our results say the points also reflect the *stopping and selection policy*: where the in-domain signal and the transfer optimum diverge, a downstream-blind protocol ships a checkpoint well past the transfer peak, and at the diagnostic level the 4k–16k cost of doing so (up to 4.7 points) exceeds what the added data buys. The matched-compute slice of Section 4.4 is the sharpest evidence that the data itself is not at fault: read at an equal step count, before the protocols diverge, more data monotonically helps. The practical reading, stated tentatively: in this regime, what a practitioner measures to stop and select is as consequential as how much unlabeled data they have. We would not have

seen this with downstream-based selection — selecting on the downstream signal hides the divergence by construction, at the price of tuning the shipped checkpoint to the evaluation task; the downstream-selected campaign in Section 4.4 gives a budget-confounded estimate of that tuning from the other side.

Does the family-level robustness ranking apply here?

The report that joint-embedding methods need more pre-training data than masked-prediction methods rests on fine-tuning evaluation [8], and the same study finds denoising autoencoders comparatively weak under linear probing. Whether the ranking carries over to linear-probe evaluation at our scale is therefore open. Our curve contributes one joint-embedding data point measured under a linear probe, at set sizes one to two orders of magnitude below those tested there — with the added nuance that at small scale, the measured “robustness to set size” depends on the selection policy as much as on the method. The five-method group curve this study is part of includes a masked-autoencoder arm under the same protocol and will offer a first side-by-side look, with the caveat that per-method tuning differs across students.

Why augmentations stayed fixed. If augmentations act chiefly as artificial dataset enlargement [20], strengthening them is the obvious lever exactly where data is scarce, and it is fair to ask why we left them alone. An augmentation is a commitment: a joint-embedding method learns to be invariant to whatever its augmentations vary [27], so a stronger augmentation buys small- N robustness by discarding the distinctions it augments away — a representation made invariant to color cannot separate red from yellow cars [27] — and which distinctions a downstream task needs is unknown at pre-training time in a pre-train-once, probe-anywhere setting. Augmentation strength also interacts with N itself, so varying both would confound the curve. Varying a capacity knob instead — the Barlow Twins projector width — leaves data, views, and objective untouched; that study is scoped as the immediate next step (Section 8).

Compute. The eight stable phases of the campaign total 10.2 GPU-hours on one consumer GPU; each cooldown adds minutes (up to roughly 25 for the largest split, sized as 20% of its selected step), and each linear probe takes about 6 minutes. We report these numbers so that others can budget similar studies.

6 Limitations

No convergence claim. Every number reported is the accuracy of the best-validation checkpoint within a fixed budget, not a best-achievable accuracy. The budgets are sized to be generously past the validation peak, and we disclose per set size whether the validation signal plateaued within its budget (Section 4.2); for any split where it did not, the curve point is a lower bound on what longer training might reach.

Selection and evaluation peek at curves. Two protocol choices inflate absolute numbers. Both are shared group protocol, so we keep them and account for them here. The downstream metric is the best test top-1 across the 100 probe epochs, which selects on the test curve; the shared evaluation script does not yet persist the final-epoch accuracy, so

the gap is unquantified in this draft and reporting it is planned for the final version. The shipped checkpoint is the maximum of a smoothed validation signal, and such a maximum is an upward-biased estimate [3]; smoothing reduces the bias but does not eliminate it, and we do not quantify what remains. Both effects inflate every curve point in the same direction. Their magnitude need not be identical across N , so they are a caveat on the curve’s exact shape as well as on absolute values.

The validation probe is sparse. The in-domain selection signal is a 200-way k NN classification with 25 gallery images per class — far sparser than published k NN protocols (DINO’s ImageNet gallery has roughly 1 300 per class [2]). Its absolute accuracy is accordingly low, and we use it only as a relative signal for ranking checkpoints within a run; we did not validate that this regime ranks checkpoints as a denser probe would.

One backbone, one transfer pair, one evaluation regime. All claims are scoped to ViT-Tiny/8 pre-trained on Tiny-ImageNet and probed linearly on CIFAR-10. The backbone is fixed by the shared group protocol, which controls it as a variable but means no claim extends to other architectures or sizes. Fine-tuning — the regime under which the family-level data-robustness ranking was established [8] — is not evaluated, and a single downstream task cannot show whether the learned invariances help or hurt elsewhere; a multi-task probe suite is future work.

One completed seed; no data resampling. The numbers in this draft come from a single pre-training seed (42); seeds 43 and 44 are in progress, and every cross-split difference smaller than a typical seed band — including the 1.8-point dip itself — should be read with that reservation (in the earlier three-seed campaign, per-split seed ranges spanned 0.5 to 2.9 points). The splits are also never resampled: the nesting attributes curve differences to N alone, but the spread will not capture sensitivity to *which* images a small split contains. The decoupling analysis of Section 4.4 rests on the k NN diagnostic, not the linear probe; the probe-level oracle comparison is pending. Finally, the campaign spans three minor revisions of the training code (a diagnostic checkpoint-saving addition entered mid-campaign, uncommitted for the three largest splits); the diff does not touch the training computation, but strict single-commit provenance holds only per split, not across the curve.

Transferred schedule evidence and a-priori choices. The WSD-schedule findings we build on — constant-rate competitiveness, cooldown shape and length — come from language-model pre-training at scales far above ours [11, 14]. Their transfer to Barlow Twins on 1k–100k images is an assumption of this study. Several recipe constants ($\lambda = 10^{-2}$, projector width 1024, crop lower bound 0.25, disabled blur, 500 warmup steps) were set a priori and not ablated. Finally, absolute accuracies shifted slightly under a framework-and-GPU change during development — the random-initialization floor moved by about 1.2 points — so all reported comparisons use numbers from a single software and hardware environment.

7 Responsible Research

Reproducibility. Every pre-training run is indexed by code commit, split size, and seed. Each checkpoint stores the git revision that produced it, and resuming a run across code versions triggers an explicit warning, so no reported number mixes two versions of the training code. The data pipeline is deterministic given its seed: the nested splits are prefixes of one published shuffled index list, and training fixes all framework seeds with deterministic cuDNN kernels. One precision exception is disclosed — pre-training matrix multiplications use TF32, which is reproducible on the same hardware but not bit-identical to fp32 across machines; the linear-probe evaluation runs in fp32 — and Section 6 discloses a cross-revision provenance caveat for the largest splits. All hyperparameters appear in Sections 3 and 4.1, including deviations from published defaults. The code, split indices, evaluation scripts, and figure scripts are kept under version control alongside the logged metrics of every run, and one consumer GPU reproduces any single curve point in hours (Section 5).

Honest reporting. Checkpoint selection never sees the downstream dataset, result claims are scoped to best-within-budget readings, and the known metric inflations and untested constants are collected in Section 6 with their direction stated. Interpretations are labeled as interpretations, and unflattering outcomes — a curve that dips, splits whose validation had not plateaued, an inflation we cannot yet quantify — are reported as such.

Data and ethics. The study uses two long-established public research datasets, Tiny-ImageNet (an ImageNet derivative) and CIFAR-10, under their existing research terms. No new data is collected and no human subjects are involved. The datasets inherit the documented limitations of their sources, including category and imagery biases; nothing in this study audits or corrects them, and the learned encoders are research artifacts for measuring data efficiency, not models intended for deployment. The compute footprint is deliberately small — a single consumer GPU for hours rather than GPU-years. Keeping the study reproducible at that scale is part of the research setting itself: making the behavior of self-supervised methods measurable with resources most groups have.

8 Conclusions and Future Work

We asked how much downstream accuracy each additional unlabeled pre-training image buys a Barlow Twins-trained small vision transformer, and what happens when the images are few. The answer, for one seed within fixed generous budgets: every amount of pre-training data helps substantially over a random-initialized encoder (by 19.9 points at 1k images, 30.0 at 100k), but the value of *additional* data is not monotone — the 4k and 8k subsets transfer worse than 1k. The smallest split over-trains, losing 18% of its peak validation accuracy with no sign of dimensional collapse, and the curve’s dip coincides with the largest measured divergence between the in-domain selection signal and downstream accuracy. The honest takeaway is that at this scale, a data-efficiency curve is a property of data *and* protocol: at the level of our diagnostic, the stopping and selection policy left

as strong a mark on the mid-range points as the data quantity did.

Three steps follow directly. The remaining pre-training seeds and the probe-level oracle comparison (the pipeline now saves the diagnostic-peak checkpoints) will show how much of the dip survives seed variance and carries from the diagnostic to the linear probe; re-running one mid-range split under a shorter budget would test the budget side of the same coin. The projector-width study — the Barlow Twins-specific capacity knob, held fixed here — is the scoped next extension. Beyond those, a multi-task probe suite would test whether the conclusions extend past CIFAR-10, and evaluating parameter-efficient fine-tuning would connect this characterization to the regime where the family-level data-robustness evidence was established [8].

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